



ROUTE OPT: A DATA- CLEANING AND APPROXIMATION ENGINE FOR CAPACITATED MULTI- VEHICLE ROUTE OPTIMIZATION WITH TIME WINDOWS

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ABSTRACT

The problem addressed is the Capacitated Vehicle Routing Problem with Time Windows (CVRPTW), which is NP-hard and difficult to solve exactly for large fleets.

Real-world logistics require fast approximation methods rather than exact solvers.

- The project introduces RouteOpt, a routing engine that:
 - Ingests raw stop and address data
 - Cleans, standardizes, and geocodes messy addresses
 - Deduplicates repeated stops
 - Clusters and assigns stops to vehicles while respecting capacity constraints
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INTRODUCTION

The core topic is the Vehicle Routing Problem (VRP) — a generalization of the Traveling Salesman Problem to multiple vehicles.

VRP is central to combinatorial optimization and logistics planning.

- Practical variants include:
 - CVRP — adds vehicle capacity limits.
 - CVRPTW — adds delivery time windows for customers.

These variants are widely used in delivery, field-service, and transportation operations.

Exact optimization methods (branch-and-bound, integer linear programming) guarantee optimal solutions but do not scale beyond a few dozen stops.

- Industry systems rely on heuristics and metaheuristics to get near-optimal solutions quickly:
 - Constructive heuristics: nearest-neighbor, Clarke-Wright savings.
 - Local/metaheuristic search: 2-opt, Or-opt, simulated annealing, tabu search.
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LITERATURE SURVEY

- **Vehicle Routing Problem (VRP):** Research mainly focuses on finding efficient routes for vehicles while serving multiple customers.
 - **Capacitated VRP (CVRP):** Existing methods consider vehicle capacity constraints to ensure that vehicle load does not exceed its maximum capacity.
 - **Time Window Constraints:** VRPTW approaches ensure that customers are served within their specified time windows, making route planning more realistic.
 - **Heuristic & Approximation Methods:** Techniques such as Greedy, 2-opt, Genetic Algorithms, and Ant Colony Optimization are used to obtain good solutions with less computation time.
 - **Research Gap:** Many existing systems focus mainly on route optimization, while RouteOpt integrates data cleaning, validation, capacity constraints, time windows, and approximate multi-vehicle routing into a unified system.
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PROBLEM STATEMENT AND OBJECTIVES

- Real-world delivery datasets often contain **missing, duplicate, inconsistent, or invalid data**, which can affect route planning.
 - Planning routes for **multiple vehicles** becomes complex when vehicle capacity constraints must be satisfied.
 - Customers may have specific **time windows**, requiring vehicles to arrive within a given time period.
 - Traditional exact optimization methods can become **computationally expensive** for large numbers of customers and vehicles.
 - There is a need for a system that **cleans and validates input data** before performing route optimization.
 - Therefore, **RouteOpt** aims to provide an efficient approximation engine that generates **feasible and optimized multi-vehicle routes** while considering **vehicle capacity and customer time-window constraints**.
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SOFTWARE AND HARDWARE REQUIREMENTS

- **Software Requirements**

- **Operating System:** Windows 10/11, macOS, or Linux.
- **Programming Language:** Python 3.x.
- **Development Environment:** Visual Studio Code, PyCharm, or Jupyter Notebook.
- **Libraries/Frameworks:** Pandas, NumPy, and suitable optimization libraries such as OR-Tools.

- **Hardware Requirements**

- **Processor:** Intel Core i5 / Apple M1 or equivalent and above.
 - **RAM:** Minimum **8 GB**.
 - **Storage:** Minimum **10 GB** of free space.
 - **Display:** 1366 × 768 resolution or higher.
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RESULTS

- **Cleaned Input Data:** The system successfully identifies and handles missing, duplicate, inconsistent, and invalid routing data.
 - **Feasible Routes:** RouteOpt generates routes that satisfy **vehicle capacity constraints** and **customer time windows**.
 - **Multi-Vehicle Optimization:** Customers are efficiently distributed among multiple vehicles based on demand and route constraints.
 - **Reduced Travel Cost:** The approximation approach helps reduce unnecessary distance and travel time compared with unoptimized routing.
 - **Faster Processing:** Approximation and heuristic techniques provide solutions faster than computationally expensive exact methods for larger datasets.
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CONCLUSION

- **RouteOpt** provides an integrated solution for data cleaning and multi-vehicle route optimization.
 - The system handles **vehicle capacity constraints and customer time windows** to generate feasible routes.
 - **Data cleaning and validation** improve the accuracy and reliability of the optimization process.
 - Approximation and heuristic techniques help generate **efficient routes with reduced computation time**.
 - The proposed system can be useful in **logistics, delivery services, transportation, and fleet management**.
 - Overall, **RouteOpt** provides a practical, scalable, and efficient approach to real-world route optimization problems.
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FUTURE SCOPE

- **Real-Time Route Optimization:** Integrate live traffic and GPS data to dynamically update routes.
 - **Machine Learning Integration:** Use machine learning to predict delivery times, traffic conditions, and customer demand.
 - **Advanced Optimization:** Implement advanced metaheuristic and hybrid algorithms to improve route quality.
 - **Real-Time Data Cleaning:** Automatically detect and correct incorrect or missing routing data during operation.
 - **Map Integration:** Integrate Google Maps, OpenStreetMap, or similar services for real-world route visualization.
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